

Assignment 3: Code Error Analysis Sheet

Programming Fundamentals (C/C++) — Debugging and Verification Log

Question Number	Error	Line Number	Error Type	Error Correction
Problem 1	Missing terminating semicolon (;) after variable initialization.	7	Syntax Error (Compilation)	Append a semicolon at the end of the declaration line: int count = 0;
Problem 2	Using assignment operator (=) instead of the equality relational operator (==) inside the control condition.	14	Logical Error (Infinite Loop)	Change the assignment to an explicit equality evaluation check: while (choice == 5)
Problem 3	Missing address-of memory reference operator (&) for input variable target in standard input stream reader function.	9	Runtime Error (Segmentation Fault)	Prepend the variable identifier with an ampersand operator to pass memory address: scanf("%d", &inputNumber);
Problem 4	Integer division precision truncation fallback during computation of fractional averages.	18	Logical Error (Precision Loss)	Force floating-point evaluation registers by explicitly dividing via a decimal literal component: average = totalSum / 5.0;
Problem 5	Missing statement control termination keyword (break;) inside a structural multi-way case branch block.	24	Logical Error (Case Fall-through)	Add an explicit break terminal command line at the absolute baseline end of the target block sequence: break;
Problem 6	Attempting to access loop counter limits or boundary elements using an completely uninitialized accumulator variable block.	5	Runtime Error (Garbage Value read)	Provide a safe initial default value assignment context to the storage target prior to entry points: int iterationIndex = 0;
Problem 7	Mismatched console output format specifier token configuration (utilizing %d instead of a float token representation).	21	Logical Display Error	Alter the format string mapping identifier to match the declared double storage type accurately: printf("Result: %.2f", calculatedValue);
Problem 8	Incorrect incremental/decremental modification expression setup leading to runaway iteration parameters.	11	Logical Error (Infinite Execution)	Invert the step tracking operator update from subtraction sequences to systematic addition logic lines: for (int i = 0; i < 10; i++)
Problem 9	Referencing an out-of-scope or entirely undeclared identifier tag label during algorithmic assignment evaluations.	32	Compilation Error	Introduce a local data type tracking reservation boundary before using the variable marker downstream:

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				double computedResult = 0.0;
Problem 10	Mismatched outer structural boundary brackets or missing terminal closing curly braces (}) at function end line.	45	Syntax Error (Compilation)	Insert a matching closing scope balance brace character at the absolute structural sequence end point to enclose code: }